Industrial Pharmacy-I
(13PH0605)

Unit-1

Preformulation Studies

Marwadi University
Marwadi Chandarana Group

NAAC A+

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SYLLABUS

- **Preformulation Studies:** Introduction to preformulation, goals and objectives, study of physicochemical characteristics of drug substances.

- **a. Physical properties:** Physical form (crystal & amorphous), particle size, shape, flow properties, solubility profile (pKa, pH, partition coefficient), polymorphism

- **b. Chemical Properties:** Hydrolysis, oxidation, reduction, racemisation, polymerisation, BCS classification of drugs & their significant

- **Application of preformulation considerations** in the development of solid, liquid, oral and parenteral dosage forms and their impact on the stability of dosage forms.

SYLLABUS

UNIT-II **10 Hours**

Tablets:

a. Introduction, ideal characteristics of tablets, classification of tablets. Excipients, Formulation of tablets, granulation methods, compression and processing problems. Equipments and tablet tooling.

b. Tablet coating: Types of coating, coating materials, formulation of coating composition, methods of coating, equipment employed and defects in coating.

c. Quality control tests: In process and finished product tests

Liquid orals: Formulation and manufacturing consideration of syrups and elixirs suspensions and emulsions; Filling and packaging; evaluation of liquid orals official in pharmacopoeia

SYLLABUS

UNIT-III **08 Hours**

Capsules:

a. ***Hard gelatin capsules:*** Introduction, Production of hard gelatin capsule shells. size of capsules, Filling, finishing and special techniques of formulation of hard gelatin capsules, manufacturing defects. In process and final product quality control tests for capsules.

b. ***Soft gelatin capsules:*** Nature of shell and capsule content, size of capsules,importance of base adsorption and minim/gram factors, production, in process and final product quality control tests. Packing, storage and stability testing of soft gelatin capsules and their applications.

Pellets: Introduction, formulation requirements, pelletization process, equipments for manufacture of pellets

UNIT-IV **10 Hours**

Parenteral Products:

a. Definition, types, advantages and limitations. Preformulation factors and essential requirements, vehicles, additives, importance of isotonicity

b. Production procedure, production facilities and controls,
aseptic processing

c. Formulation of injections, sterile powders, large volume parenterals and lyophilized products.

d. Containers and closures selection, filling and sealing of ampoules, vials and infusion fluids. Quality control tests of parenteral products.

Ophthalmic Preparations: Introduction, formulation considerations; formulation of eye drops, eye ointments and eye lotions; methods of preparation; labeling, containers;

SYLLABUS

Ophthalmic Preparations: Introduction, formulation considerations; formulation of eye drops, eye ointments and eye lotions; methods of preparation; labeling, containers; evaluation of ophthalmic preparations

UNIT-V **10 Hours**

Cosmetics: Formulation and preparation of the following cosmetic preparations: lipsticks, shampoos, cold cream and vanishing cream, tooth pastes, hair dyes and sunscreens.

Pharmaceutical Aerosols: Definition, propellants, containers, valves, types of aerosol systems; formulation and manufacture of aerosols; Evaluation of aerosols; Quality control and stability studies.

Packaging Materials Science: Materials used for packaging of pharmaceutical products, factors influencing choice of containers, legal and official requirements for containers, stability aspects of packaging materials, quality control tests.

REFERENCES

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REFERENCE/TEXTBOOKS

1. Modern Pharmaceutics by Gilbert S. Banker & C.T. Rhodes, 3rd Edition
2. Remington: The Science and Practice of Pharmacy, 20th edition Pharmaceutical Science (RPS)
3. Theory and Practice of Industrial Pharmacy by Liberman & Lachman
4. Pharmaceutical dosage forms- Tablets, volume 1-3 by H.A. Liberman, Leon Lachman & J. B. Schwartz
5. Pharmaceutical dosage form- Parenteral medication vol- 1&2 by Liberman & Lachman
6. Introduction to Pharmaceutical Dosage Forms by H. C.Ansel, Lea &Febiger, Philadelphia, 5th edition, 2005

Introduction: Preformulation Studies

Preformulation Studies

1. Introduction to Preformulation

- Preformulation is the **initial step in the rational development** of dosage forms of a drug substance. It involves the **characterization of the physical, chemical, and mechanical properties** of a drug molecule. These studies help in understanding the nature of the drug and identifying **challenges related to formulation** and stability before proceeding to actual dosage form development.
- Preformulation bridges the gap between drug discovery and formulation development. It aids in selecting the right form of the drug, suitable excipients, and compatible dosage forms, ultimately leading to better **bioavailability, stability, and manufacturability**.

Introduction: Preformulation Studies

❑ Need of Preformulation study

- Preformulation studies are an important foundation in the development of both API as well as drug Products.
- They directly or indirectly influence:
- Selection of drug candidate
- Selection of formulation components
- API & drug product Manufacturing
- Determination of the most appropriate container/closure system
- Development of analytical method
- Toxicological Strategic Management process

Introduction: Preformulation Studies

2. Goals and Objectives of Preformulation Studies

Goals:

To **characterize the physicochemical properties** of the drug molecule.

To identify **potential problems** associated with the formulation process.

To design a **stable, safe, and effective dosage form**.

To improve the **biopharmaceutical properties** of the drug.

To support **regulatory documentation** by providing essential data.

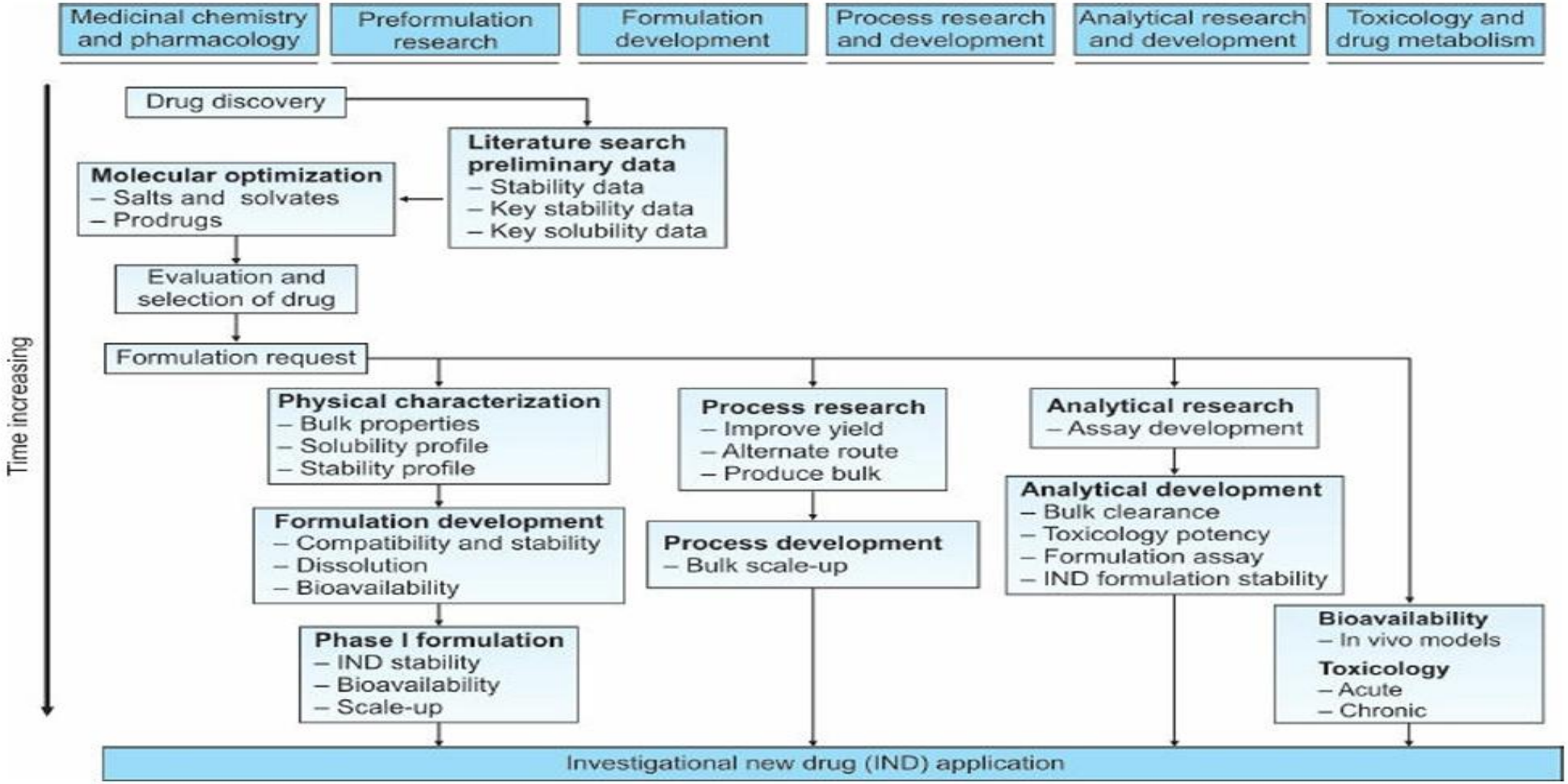
Objectives:

To determine the **physical and chemical properties** of the API (Active Pharmaceutical Ingredient).

To evaluate the **compatibility of the drug with excipients**.

To assess the **stability profile** of the drug under various environmental conditions (light, heat, humidity).

Introduction: Preformulation Studies



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Early Drug Discovery	Preclinical Studies	Clinical Development	FDA Review	Post-market Monitoring
▪ Target Identification & Validation ▪ Hit Discovery ▪ Assay Development & Screening ▪ High Throughput Screening ▪ Hit to Lead ▪ Lead Optimization	▪ In Vivo, In Vitro & Ex Vivo Assays ▪ ADME ▪ Proof of Concept ▪ Drug Delivery ▪ Formulation Optimization & Bioavailability ▪ Dose Range Finding ▪ IND-enabling Studies ▪ IND Application	▪ Phase I – Healthy Volunteer Study ▪ Phase II and Phase III – Studies in Patient Population ▪ Dose Escalation, Single Ascending & Multiple Dose Studies ▪ Safety & Efficacy ▪ Pharmacokinetic Analysis ▪ Bioanalytical Method Development and Validation	▪ NDA / ANDA / BLA Application ▪ FDA Approval ▪ Drug Registration	▪ FDA Adverse Event Reporting System (FAERS)

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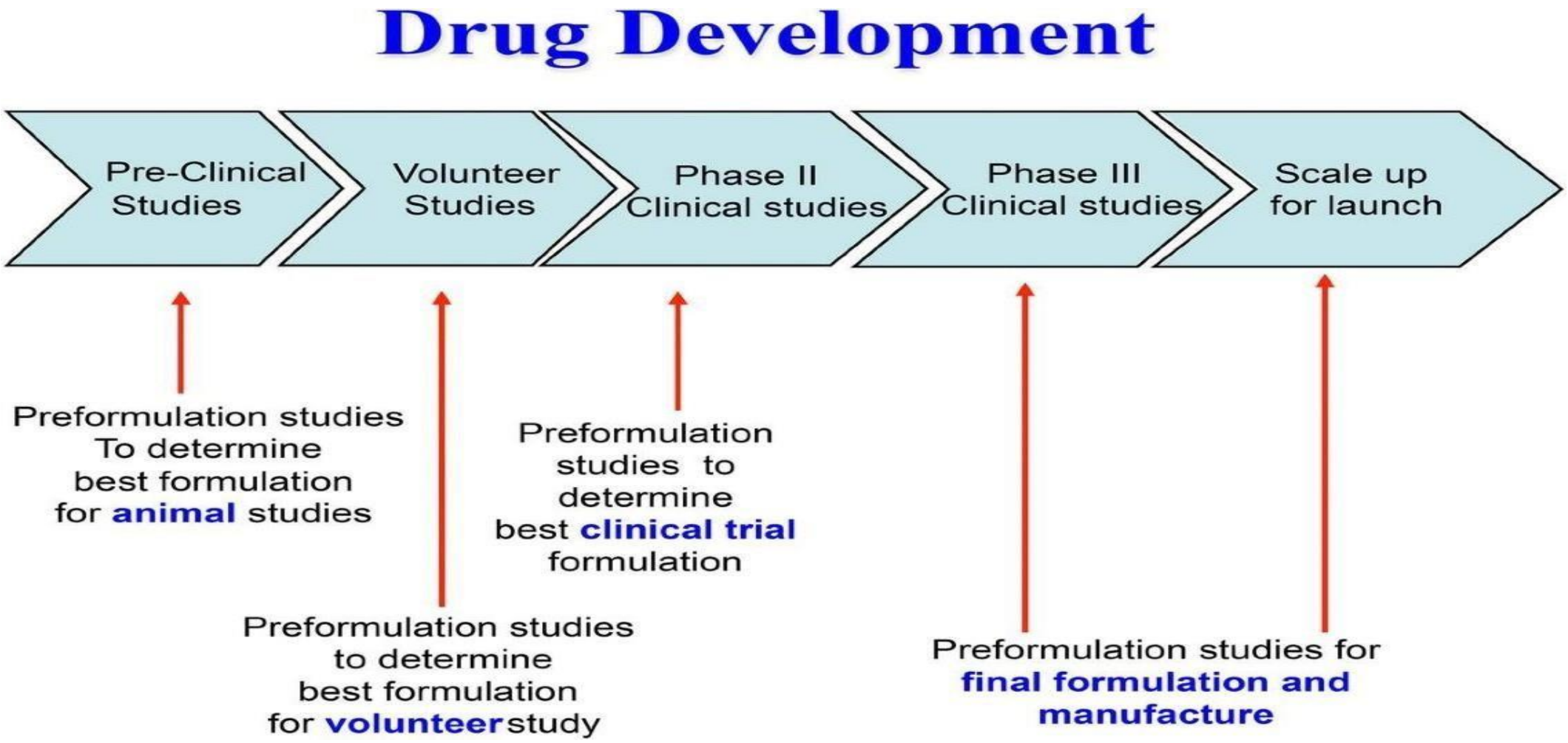

Introduction: Preformulation Studies

1. Physicochemical Properties:

Organoleptic characteristics
Particle Size
Particle
Shape
Surface
area
Purity

Principal areas of Preformulation Research

2. Bulk Characterization

- Crystallinity and Polymorphism
- Hygroscopicity
- Micromeritic Properties
  - ✓ Particle Characterization
  - ✓ Density and Porosity
  - ✓ Powder Flow Properties

3. Solubility Studies

•pKa Determinations
•pH Solubility Profile and Common Ion Effects
•Effect of Temperature
•Solubilization
•Partition Coefficient
•Dissolution

4. Stability Analysis

•Solution Stability
•Solid-state Stability
•Drug-excipient Compatibility

Introduction: Preformulation Studies

Study of Physicochemical Characteristics of Drug Substances

Understanding the physicochemical properties of a drug is vital as it affects **solubility, stability, bioavailability**, and processability.

A. Physical Properties

Organoleptic Properties

Color, odor, taste, and appearance.

Important for patient acceptability and product aesthetics.

Introduction: Preformulation Studies

Particle Size and Shape

Affects solubility, dissolution rate, and bioavailability.

Also influences flowability and compressibility.

Crystalline and Amorphous Nature

Crystalline drugs are more stable but less soluble.

Amorphous forms show higher solubility but may be less stable.

Introduction: Preformulation Studies

B. Chemical Properties

Solubility Analysis

Solubility in water, buffers, and organic solvents.

Critical for bioavailability and formulation approach.

pKa and Dissociation Constant

Indicates the degree of ionization at different pH values.

Affects solubility, absorption, and formulation pH.

Introduction: Preformulation Studies

B. Chemical Properties

Partition Coefficient (Log P)

Distribution of drug between lipophilic and hydrophilic phases.

Indicates lipophilicity, permeability, and potential for absorption.

Stability Studies

Thermal, oxidative, photolytic, and hydrolytic degradation.

Helps in identifying degradation pathways and storage conditions.

Introduction: Preformulation Studies

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1. Preformulation studies are primarily carried out to:

A. Develop a marketing strategy for a drug
B. Establish the pharmacological action of a drug
C. Investigate the physicochemical properties of a drug prior to formulation
D. Conduct clinical trials of the drug

2. Which of the following is NOT an objective of preformulation studies?

A. To ensure stability of the drug
B. To select a suitable dosage form
C. To determine therapeutic index
D. To improve bioavailability

3. **Which physicochemical property of a drug most directly affects its dissolution rate?**

A. Melting point
B. Particle size
C. Polymorphism
D. Crystal habit

Introduction: Preformulation Studies

M.C.Q

4. The study of polymorphism during preformulation is important because it influences:

A. Drug color
B. Manufacturing cost
C. Solubility and bioavailability
D. Packaging material

5. **Which parameter is most important for predicting the absorption of a drug across biological membranes?**

A. Molecular weight
B. Partition coefficient
C. Bulk density
D. Flow property

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Preformulation Studies: Physical Properties

1. Physical Form (Crystalline and Amorphous)

Crystalline Form:

Molecules are arranged in an **ordered, repeating pattern**.

Characterized by **definite melting point** and **higher stability**.

Usually has **lower solubility** and **slower dissolution rate**.

Crystals can exhibit polymorphism (exist in different crystal forms).

Advantages:

Greater physical and chemical stability.

Ease of purification due to sharp melting point.

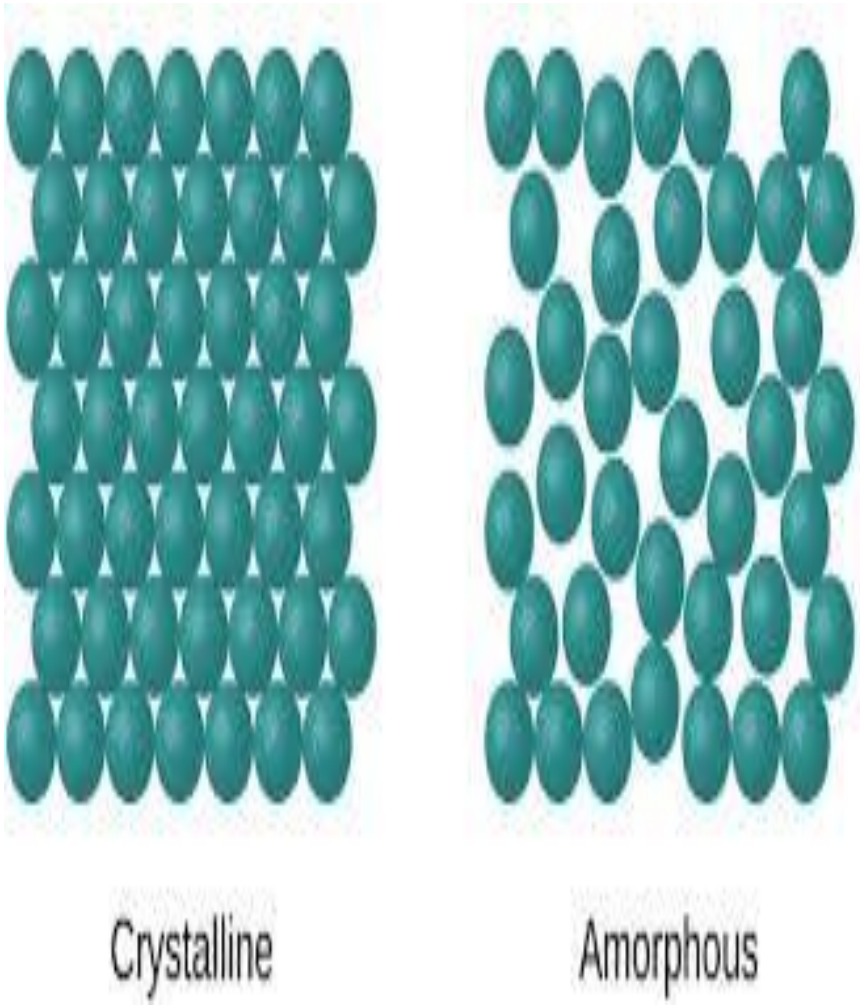
Preformulation Studies: Physical Properties

Amorphous Form:

- Molecules are in a **disordered, random arrangement**.
- No definite melting point; shows **glass transition temperature (Tg)**.
- Typically has **higher solubility** and **faster dissolution**.
- Less stable; more prone to **degradation** and **recrystallization**.

Use in formulation:

- Amorphous forms are used to improve bioavailability of poorly soluble drugs.



Preformulation Studies: Physical Properties

Crystalline	Amorphous
Crystalline form have fixed internal structure.	Amorphous forms do not have any fixed internal structure
Crystalline form has lesser thermodynamic energy as compared to its amorphous form.	Amorphous form has higher thermodynamics energy than its crystalline form
Crystalline forms are more stable than its amorphous forms.	Amorphous forms are less stable than its crystalline forms
Crystalline forms has lesser solubility than its amorphous form	Amorphous forms have a greater solubility than its crystalline forms
Crystalline forms has less tendancy to change its form during storage	Amorphous tends to the word to more stable form during storage

Preformulation Studies: Physical Properties

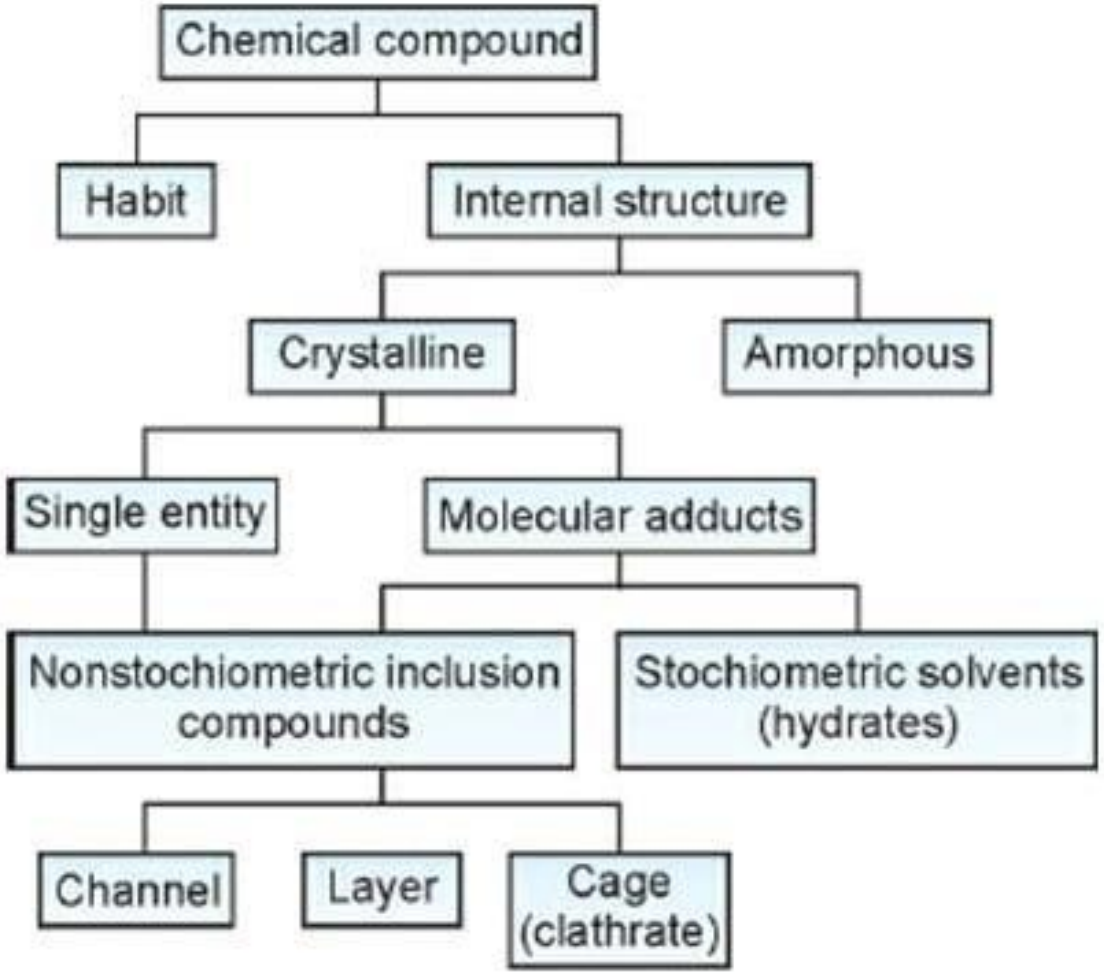


Fig. 9.13: Outline of differentiating habit and crystal chemistry of a compound

Preformulation Studies: Physical Properties

- Crystal habit & internal structure of drug can affect bulk & physicochemical property of molecule.
- **Crystal habit** is description of outer appearance of crystal.
- **Internal structure** is molecular arrangement within the solid.
- Change with internal structure usually alters crystal habit.
- Eg. Conversion of sodium salt to its free acid form produce both change in internal structure & crystal habit

Preformulation Studies: Physical Properties

- The **degree of crystallinity** of drug substance has marked effect on its hardness, density, transparency and diffusion
- This influences both the choices of delivery system and activity as determined by the rate of delivery
- As **crystalline compound contain stoichiometric or non stoichiometric crystallization** Solvent
- **Non stoichiometric adducts are called inclusion or clathrates**
- While **stoichiometric adducts** are called **solvates**
- The crystalline form of penicillin G as a potassium or sodium salt is considerably more stable and result in excellent therapeutic response than amorphous forms.
- If the compound contain water as a solvent then it is known as **hydrate**

Preformulation Studies: Physical Properties

Stoichiometric Adducts

Definition:

Stoichiometric adducts are solid complexes formed between a drug and another substance in a **fixed, definite molar ratio**.

Characteristics:

•Definite chemical composition

•Formed through **specific chemical interactions** (e.g., hydrogen bonding, ionic interaction)

•Can be represented by a chemical formula

•Show reproducible physicochemical properties

Preformulation Studies: Physical Properties

Examples:

•Drug–solvent hydrates (e.g., caffeine hydrate)

•Drug–coformer complexes

•Salt formation (e.g., drug–acid or drug–base salts)

Importance in Preformulation:

•Affect **solubility**, **stability**, and **bioavailability**

•Help in **improving dissolution rate**

•Useful in selecting suitable **dosage forms**

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2. Non-Stoichiometric Adducts

Definition:

Non-stoichiometric adducts are complexes where the drug and another substance combine in **variable or indefinite ratios**, without a fixed molecular composition.

Characteristics:

- No definite chemical formula
- Interaction is **physical rather than chemical**
- Composition may vary with conditions such as temperature or concentration
- Generally reversible in nature

Preformulation Studies: Physical Properties

Examples:

- Adsorption of drugs on excipients
- Drug–polymer interactions
- Solid dispersions
- Inclusion complexes with variable guest content

Importance in Preformulation:

- Influence **drug release**, **stability**, and **bioavailability**
- Important in **controlled and sustained release formulations**
- Affect **drug–excipient compatibility**

Preformulation Studies: Physical Properties

Non-Stoichiometric inclusion compounds (or adducts)

• In these crystals solvent molecules are entrapped within the crystal lattice and the number of solvent molecules are not included in stoichiometric number.

Depending on the shape they are of three types :-

- **Channel:** When the crystal contains continuous channels in which the solvent molecule can be included. e.g . Urea forms channel.
- **Layers**:- Here solvent molecules are entrapped in between layers of crystals.
- **Clathrates(Cage):-** Solvent molecules are entrapped within the cavity of the crystal from all sides.

Preformulation Studies: Physical Properties

Stoichiometric inclusion compounds (or stoichiometric adducts)

- This molecular complex has incorporated the **crystallizing solvent molecules** into specific sites within the **crystal lattice and has stoichiometric number of solvent molecules complexed**
- When the **incorporated solvent is water**, the complex is called **hydrates and when the** solvent is **other than water**, the complex is called **solvates**.

Preformulation Studies: Physical Properties

Depending on the ratio of water molecules within a complex the following nomenclature is followed.

- **Anhydrous** : 1 mole compound + 0 mole water
- **Hemihydrate**: 1 mole compound + ½ mole water
- **Monohydrate**: 1 mole compound + 1 mole water
- **Dihydrate** : 1 mole compound + 2 moles water

Preformulation Studies: Physical Properties

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Preformulation Studies: Physical Properties

2. Particle Size and Shape

Particle Size:

Affects **dissolution rate**, **bioavailability**, **content uniformity**, **stability**, and **flow properties**.

Smaller particles → larger surface area → faster dissolution.

Very fine particles may cause **poor flow**, **electrostatic charges**, and **agglomeration**.

Measured by:

Sieving, microscopy, laser diffraction, sedimentation, Coulter counter.

Preformulation Studies: Physical Properties

Particle Shape:

Includes **spherical, needle-shaped, plate-like**, etc.

Affects **flow, packing density**, and **surface contact**.

Spherical particles → better flow and compressibility.

Irregular shapes → increased surface area but poor flow.

Preformulation Studies: Physical Properties

- Particle size represents the dimensions of solid powders, liquid particles, and gas bubbles.
- Particle size mainly influences dissolution rate.
- The absorption rate of poorly soluble drugs can be improved by reducing the particle size and thus enhances the bioavailability of drugs.
- Particle size is also considered during the development of new chemical entities (NCEs) as it affects the physical properties of active pharmaceutical ingredients and excipients.
- Particle size and shape significantly affect various manufacturing steps, such as mixing, granulation, drying, milling, blending, coating, encapsulation, and compression

Preformulation Studies: Physical Properties

- **Very coarse (No. 8):** All particles pass through a No. 8 sieve and not more than 20% pass through a No.60 sieve.
- **Coarse (No. 20):** All particles pass through a No. 20 sieve and not more than 40% pass through a No.60 sieve.
- **Moderately coarse (No. 40):** All particles pass through a No. 40 sieve and not more than 40% pass through a No.80 sieve.
- **Fine (No.60):** All particles pass through a No.60 sieve and not more than 40% pass through a No.100 sieve.
- **Very fine (No. 80):** All particles pass through a No. 80 sieve. There is no limit to greater fineness.

Preformulation Studies: Physical Properties

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Particle size can influence a variety of important factors, including the following

- **Dissolution rate** of particles intended to dissolve; drug micronization can increase the rate of drug dissolution and its bioavailability.

- **Suspendability** of particles intended to remain undissolved but uniformly dispersed in a liquid vehicle (e.g., fine dispersions have particles approximately 0.5 to 10 μm)

- **Uniform distribution** of a drug substance in a powder mixture or solid dosage form to ensure dose-to-dose content uniformity.

Preformulation Studies: Physical Properties

- **Penetrability** of particles intended to be inhaled for deposition deep in the respiratory tract (e.g., 1 to 5 µm).

- **Lack of grittiness** of solid particles in dermal ointments, creams, and ophthalmic preparations (e.g.,fine powders may be 50 to 100 µm in size).

- Very fine materials are difficult to handle, overcome by creating solid solution in a carrier (water- soluble polymer).
- Important to decide, maintain, and control a desired size range
- Drawbacks to grinding: - material losses - static charge build-up - Aggregation

=> increase hydrophobicity

=> lowering dissolution rate - polymorphic or chemical transformations

Preformulation Studies: Physical Properties

Size reduction not done when:

- When drug is unstable.
- Degrade in solution form.
- Produce undesirable effects.
- When sustained effect is desired.

Particle size determination (PSD) - Methods

1. Optical Microscopy (Counting)
2. Sieving Method (Separation)
3. Sedimentation Method
4. Conductivity Method

Preformulation Studies: Physical Properties

1. Organoleptic characteristics of a drug substance include:

A. Solubility and dissolution rate

B. Taste, color, and odor

C. Particle size distribution

D. Chemical stability

2. **Reduction in particle size of a drug mainly results in:**

A. Decreased surface area

B. Increased dissolution rate

C. Reduced bioavailability

D. Increased chemical degradation

3. **Particle shape of a drug substance primarily influences:**

A. Drug taste

B. Flow properties and packing behavior

C. Chemical purity

D. Melting point

Preformulation Studies: Physical Properties

4. An increase in surface area of drug particles will generally lead to:

A. Decreased dissolution rate

B. Increased stability

C. Increased dissolution and absorption

D. Reduced bioavailability

5. **Purity of a drug substance is most important because impurities may:**

A. Improve drug solubility

B. Enhance therapeutic action

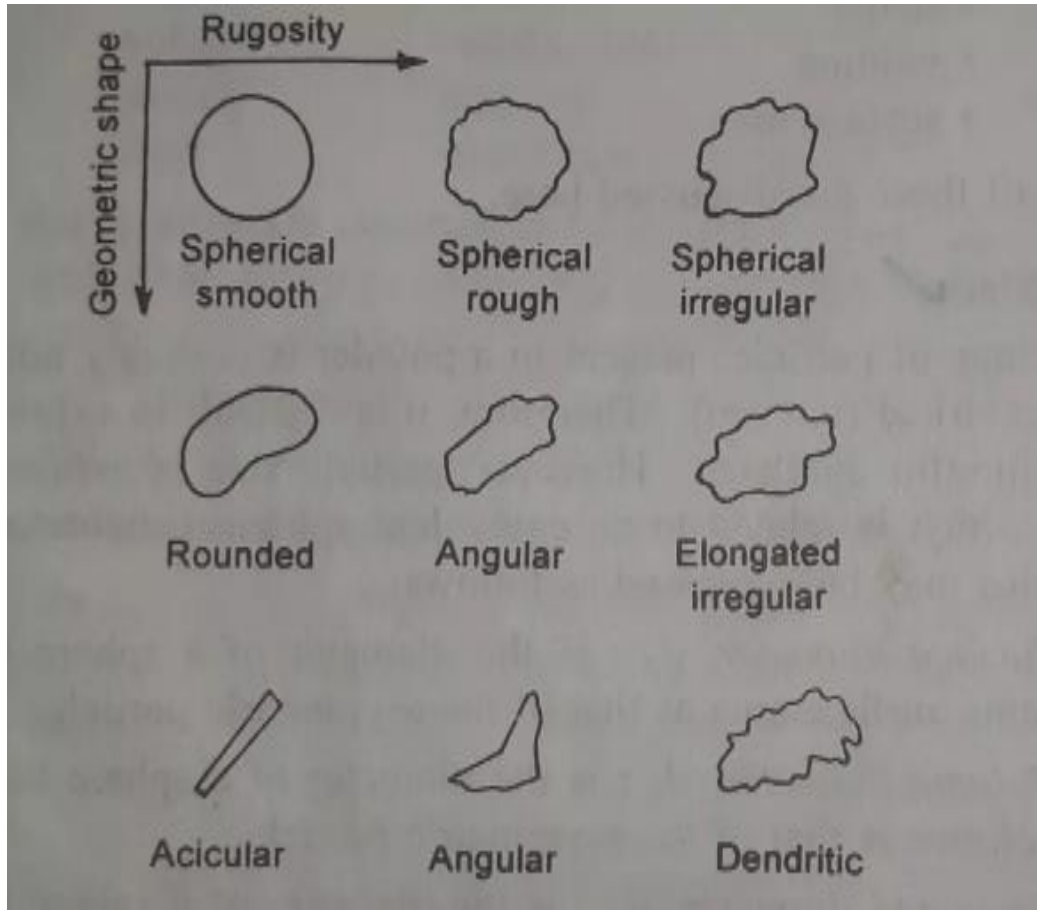
C. Cause toxicity or instability

D. Increase flow property

2. Particle Shape

According to their shape particle divided are into two groups.

1) Symmetrical Particle
2) Asymmetrical Particle



2. Particle Shape

Particle Shape:

- A **sphere** has a minimum surface area per unit volume.
- The **more asymmetric the particle** becomes, the **greater is the surface area** per unit volume.
- Hence, it may be necessary to determine the extent of asymmetry in a particle.
- A sphere is characterized by its diameter. An asymmetric particle is more difficult to characterize in terms of surface diameter.
- Hence, the asymmetric particle's surface diameter is measured in terms of some equivalent spherical diameter.

3. Polymorphism

It is the ability of the compound to crystallize as more than one distinct crystalline species with different internal lattice.

Different crystalline forms are called polymorphs but their chemical composition remain same.

Polymorphs are of 2 types

- Enatiotropic
- Monotropic

The polymorph which can be changed from one form into another by varying temp or pressure is called as **Enantiotropic polymorph**.

- **Eg.** Sulphur.

One polymorph which is unstable at all temp. & pressure is called as **Monotropic polymorph**.

- **Eg.** Glyceryl stearate.

Polymorphs differ from each other with respect to their physical property such as

- Solubility
- Melting point
- Density
- Hardness
- Compression characteristic

Characteristics of polymorphs

Characteristics	Stable polymorph	Metastable polymorph	Unstable polymorph
Packing of molecules in crystal lattice	Tightly packed	Less tightly packed	Loosely packed
Melting point	Highest	Moderate	Lowest
Rate of dissolution	Lowest	Moderate	Highest

- Depending upon their relative stability, one of the several polymorphic form will be physically more stable than others.
- Stable polymorph represents the lowest energy state, has highest melting point and least aqueous solubility.
- Metastable form represent the higher energy state, have lower melting point and high aqueous solubility .
- Metastable form converted to the stable form due to their higher energy state.
- Metastable form shows better bioavailability and therefore preferred in formulations.

- Only 10% of the pharmaceuticals are present in their metastable form.
- Solubility (particularly important in suspensions and biopharmaceutically), melting point, density, crystal shape, optical and electrical properties and vapour pressure are often very different for each polymorph.
- Polymorphism is remarkably common, particularly within certain structural groups: 63% of barbiturates, 67% of steroids and 40% of sulphonamides exhibit polymorphism.
- The steroid progesterone has five polymorphs, whereas the sulphonamide sulphabenzamide has four polymorphs and three solvates.

- Many drugs are hydrophobic and have very limited solubility in water. If the drug remains in several polymorphic forms then the stable one will produce the slowest rate of dissolution and it may show minimum bioavailability.
- For highly water soluble drugs polymorphism does not show any problem in dissolution rate
- Example: Chloramphenicol palmitate has three polymorphs α(stable), β(metastable) and γ (unstable).
- When chloramphenicol palmitate suspension is prepared from α or β polymorph it is found that bioavailabilty is higher with the metastable form.

Example:

- Two polymorphs of aspirin can be obtained by recrystallization of aspirin from 95% ethanol or n hexane.
- The polymorph obtained from n-hexane is found to have greater solubility in water than the polymorph obtained from ethanol

Methods of characterization of polymorphs

1. Hot stage microscopy,
2. Differential Thermal Analysis,
3. Differential Scanning Calorimetry
4. Thermogravimetric Analysis (TGA)
5. X-ray powder diffraction
6. IR-Spectroscopy

M.C.Q

1. **Reduction in particle size of a drug mainly results in:**
   A. Decreased dissolution rate
   B. Increased chemical stability
   C. Increased surface area and dissolution rate
   D. Decreased bioavailability
2. **2. Which method is most commonly used for determining particle size of powders in the range of 1–100 µm?**
   A. Sieving method
   B. Optical microscopy
   C. Sedimentation method
   D. Laser diffraction

3. Which particle shape generally exhibits the best flow properties?
A. Needle-shaped particles
B. Flaky particles
C. Irregular particles
D. Spherical particles

4. In preformulation studies, polymorphism is important because different polymorphic forms may differ in:
A. Colour only
B. Chemical structure
C. Solubility and bioavailability
D. Molecular weight

5. Which of the following is an example of polymorphism?

A. Formation of solvates

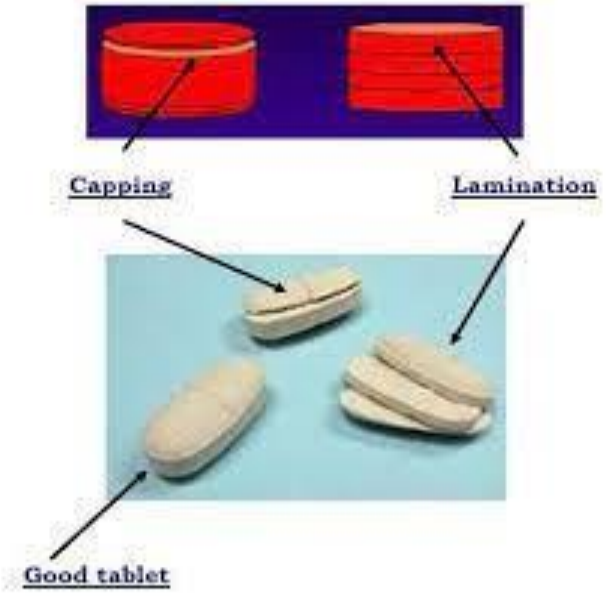
B. Same drug existing in different crystalline forms

C. Presence of impurities

D. Change in particle size

4. Flow Properties of Powders

- Powders may be free-flowing or cohesive (Sticky).
- Many common manufacturing problems are attributes to powder flow when powder transfer through large equipment such as hopper.
- Uneven powder flow -excess entrapped air within powders capping or lamination.
- Uneven powder flow increase particle's friction with die wall causing lubrication problems and increase dust contamination risks during powder transfer.



- Powder storage, which for example result in caking tendencies within a vial or bag after shipping or storage time.

- Separation of small quantity of the powder from the bulk-specifically just before the creation of individual doses such as during tableting, encapsulation and vial filling which affect the weight uniformity of the dose (under or over dosage).

Parameters to evaluate the flowability of a powder.

- Carr's compressibility index.
- Hausner ratio.
- The angle of repose(Θ).

1. **Bulk Density (ρ_β):** Bulk density is defined as the **mass of a powder divided by the bulk volume it occupies**, including the volume of **inter-particle voids**. The powder is measured **without applying any external force**.

Formula:

$$\text{Bulk density } (\rho_\beta) = \frac{M}{V_b}$$

Where:

- M = mass of powder
- V_b = bulk volume

2. Tapped Density (ρ_t)

Definition:

Tapped density is the **mass of a powder divided by the volume occupied after tapping** the container mechanically until **no further volume change occurs**.

Formula:

$$\text{Tapped } density\ (\rho_t) = \frac{M}{V_t}$$

Where:

- M = mass of powder
- V_t = tapped volume

3. Porosity (ε)

Definition:

Porosity is the **fraction of the total volume of a powder bed that is occupied by void spaces** (air spaces) between particles.

Formula:

$$\text{Porosity } (\varepsilon) = \frac{V_b - V_t}{V_b}$$

or, in terms of densities:

$$\text{Porosity } (\varepsilon) = 1 - \frac{\rho_b}{\rho_t}$$

100×)if expressed as percentage)

1. Carr's Compressibility Index (CI)

Definition:

Carr's Compressibility Index is a measure of the **compressibility of a powder**, which indirectly indicates its **flowability**. It is based on the difference between **bulk density** and **tapped density**.

Formula:

$$\text{Carr's } Index\ (\%) = \frac{Tapped\ density - Bulk\ density}{Tapped\ density} \times 100$$

Significance:

- Lower values indicate **better flow properties**
- Higher values indicate **poor flow and high cohesiveness**

Carr's compressibility index

- A volume of powder is filled into a graduated glass cylinder and repeatedly tapped for a known duration. The volume of powder after tapping is measure.
- Carr's index (%) $= 100\left(\frac{Tapped\ density - Bulk\ density}{Tapped\ density}\right)$
- Bulk density $= \frac{\text{Weight}}{\text{Bulk volume}}$
- Tapped density $= \frac{\text{Weight}}{\text{Tapped volume}}$

Carr's compressibility index

Relationship between powder flowability and % compressibility

Flow description	% Compressibility	Examples
Excellent flow	5 – 15	Free flowing granules
Good	12 – 16	Free flowing powdered granules
Fair to Passable	18 – 21	Powdered granules
Poor	23 – 35	Very fluid powders
Very Poor	33 -38	Fluid cohesive powders
Extremely poor	> 40	Fluid cohesive powders

2. Hausner Ratio

Definition:

Hausner ratio is an index used to evaluate the **flow characteristics of powders** by comparing the **tapped density** to the **bulk density**.

Formula:

$$\text{Hausner } Ratio = \frac{Tapped\ density}{Bulk\ density}$$

Significance:

- A **lower Hausner ratio** indicates **free-flowing powder**
- A **higher ratio** indicates **poor flowability**

Hausner Ratio

$$\text{Hausner ratio} = \frac{\text{Tapped density}}{\text{Poured or bulk density}}$$

Hausner ratio was related to inter-particlefriction:

- Value less than 1.25 indicates good flow (=20%Carr).
- The powder with low inter-particle friction, such as coarsespheres.
- Valuegreater than1.5 indicates poor flow (= 33% Carr's or CompressibilityIndex)).
- More cohesive, less free-flowing powders such as flakes.
- Between 1.25 and 1.5 (added glidant normally improves flow).
- >1.5 (added glidant doesn't improve flow).

- Significance: It is related to interparticle friction. So it can be used to predict powder flow properties.
- For coarse, free flowing powders the Hausner ratio is approximately 1.2.
- For more cohesive, less free-flowing powder (e.g. flakes) will have a Hausner ration greater than 1.5.
- Interpretation: Greater the Hausner ratio more cohesive will be the powder and flowability will be reduced.

3. Angle of Repose (θ)

The angle of repose is the **maximum angle** formed between the surface of a **pile of powder** and the horizontal plane. It is a measure of **inter-particle friction** and **powder flowability**.

Formula:

$$\tan\theta = \frac{h}{r}$$

Where:

- h = height of the powder cone
- r = radius of the base

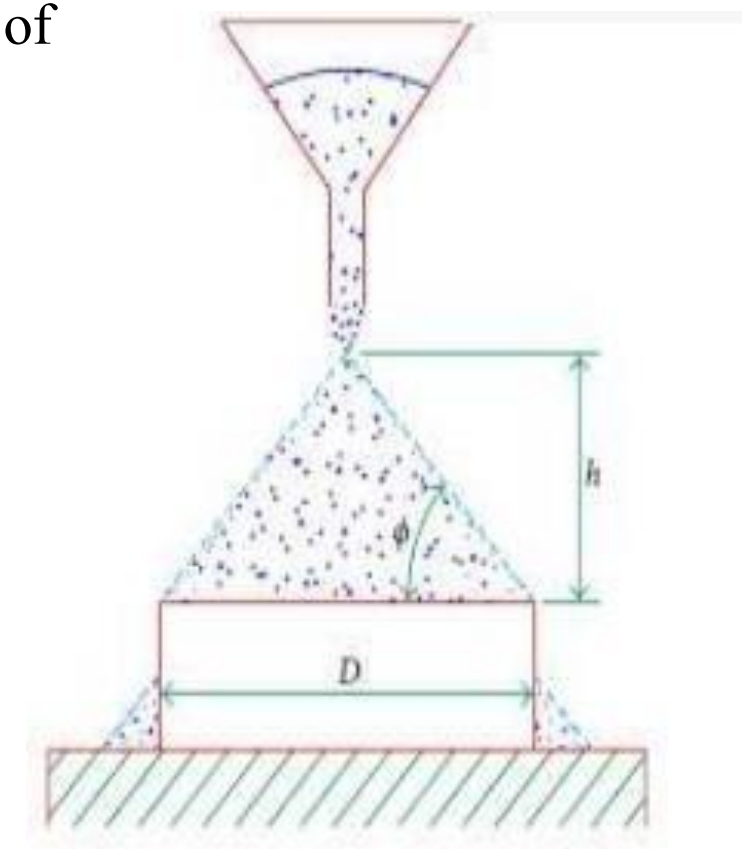
Significance:

- Smaller angle → **better flow**
- Larger angle → **poor flow**

Angle of repose

- Angle of repose of a powder sample is a parameter that shows interparticle cohesion.
- If cohesion between powder particles is less then powder flow-property is better.
- Due to cohesion, powder experiences a drag force against flow.
- Interparticle cohesive forces are due to non-specific van der Waals forces.
- It increases as particle size decreases and moisture content increases.
- Surface tension forces between adsorbed liquid layers at the particle surface and Electrostatic forces arising from contact or friction with the wall of the equipment.

- The sample is poured onto the horizontal surface and the angle of the resulting pyramid is measured.
- The user normally selects the funnel orifice through which the powder flows slowly and reasonably constantly.



Angle of repose=Tan ø=h/r

$Ø=\tan^{-1}(h/r)$

Flow Property	Angle of Repose
Excellent	25-30
Good	31-35
Fair	36-40
Passable	41-45
Poor	46-55
Very Poor	56-65
Very, Very Poor	>65

The rougher and more irregular the surface of the particles, the higher will be the angle of repose.

M.C.Q

1. Which parameter is most commonly used to evaluate the flow property of a powder?
A. Bulk density
B. Angle of repose
C. True density
D. Particle density

2. A powder showing a Carr's Compressibility Index value of 12% can be classified as:
A. Very poor flow
B. Poor flow
C. Fair flow
D. Good flow

3. Hausner ratio is calculated using which of the following parameters?
A. Bulk volume and tapped volume
B. True density and bulk density
C. Tapped density and bulk density
D. Particle size and shape

4. An increase in inter-particle friction will result in:

A. Decrease in angle of repose

B. No change in flow property

C. Increase in angle of repose

D. Improved flow

5. Which of the following angle of repose values indicates excellent flow of powder?

A. > 45°

B. 35–40°

C. 30–35°

D. < 30°

Factors affecting the flow properties of powder

1. Particle's size & Distribution
2. Particle shape & texture
3. Surface Forces

How flow properties can be improved

1. Alteration of Particle's size & Distribution
2. Alteration of Particle shape & texture
3. Alteration of Surface Forces
4. Formulation additives (Flow activators)

Alteration of Particle's size & Distribution

- There is certain particle size at which powder's flow ability is optimum.
- Coarse particles are more preferred than fine ones as they are less cohesive.
- The size distribution can also be altered to improve flow ability by removing a proportion of the fine particle fraction or by increasing the proportion of coarser particle's such as occurs in granulation.

Alteration of Particle shape & texture

- Generally, more spherical particles have better flow properties than more irregular particles.
- Spherical particles are obtained by spray drying, or by temperature cycling crystallization.

Alteration of Particle shape & texture

- Particles with very rough surfaces will be more cohesive and have a greater tendency to interlock than smooth surfaced particles.

Alteration of Surface Forces

- Reduction of electrostatic charges can improve powder flowability.
- Electrostatic charges can be reduced by altering process conditions to reduce frictional contacts.
- Moisture content of particle greatly affects powder's flowability.
- Adsorbed surface moisture films tend to increase bulk density and reduce porosity.
- Drying the particles will reduce the cohesiveness and improve the flow.
- Hygroscopic powder's stored and processed under low humidity conditions.

Addition of Formulation additives (Flow activators)

- Flow activators are commonly referred as a glidants.
- Flow activators improve the flowability of powders by reducing adhesion and cohesion.
- e. g. Talc, maize starch and magnesium stearate.

Application of free flowing powders in pharmacy

- Powders are required to produce tablets or capsules.
- Free flowing powders flow uniformly into the die cavity of tablet punching machine and inside the empty gelatin shells.
- A free flowing powder produces uniform content of drug in the tablets and capsules.
- Free flowing powders show reproducible filling of tablet dies and capsules dosators, which improve weight uniformity and physicomechanical properties (e.g. hardness).
- Poor powder flow can result in excess entrapped air within powders which in some high-speed tableting conditions may promote capping or laminations.

Application of free flowing powders in pharmacy

- Poor powder flow can result from excess fine particles in a powder, which increase friction in between particle and die wall.
- It may cause lubrication problem.
- In industry powders are required to flow from one location to another and this is achieved by different methods, such as gravity feeding, fluidization in gases and liquids and hydraulic transfer.
- In each of these examples powders are required to flow.
- In capsule filling machine, especially Lily type capsule filling machine the powder must be free flowing to uniformly fill the base of the capsule shells. In case of Zanasi-type filling machine cohesive powders are required.

THANK YOU